

Harshit Singh

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Education

Vellore Institute of Technology, Chennai — Chennai, India
B.Tech, Electronics and Computer Engineering — CGPA: 7.97 — *2023 – 2027*

New Era School (CBSE) — Ghaziabad, Uttar Pradesh
Class XII: 87.16% — Class X: 89.83% — *2020 – 2023*

Experience

AI & Robotics Intern — Autonomous Mobile Robotics [**Certificate**] — Jul 2025 – Aug 2025
STEMTEC AI & Robotics Technology Pvt. Ltd. — *Chennai, India*

- Developed a **ROS 2 Jazzy reactive navigation node** processing `/scan` LiDAR data at **10 Hz**, enabling real-time obstacle avoidance within a **0.75 m safety radius** and reducing collision events during indoor navigation tests by **∼80%**.
- Implemented timer-driven velocity control nodes publishing to `/cmd_vel` with **sub-50 ms command latency**, achieving stable motion at **0.4 m/s** and consistent bidirectional odometry validation across simulation and hardware deployments.

Projects

SPARC — Sign Language Processing & Recognition System *(OpenCV, Deep Learning)* [**Demo**] [**Web-App**]

- Developed a real-time ISL/ASL recognition pipeline (A–Z, 0–9, words) trained on **12k+ samples**, achieving **92.4% accuracy** at **15–20 FPS** with end-to-end latency under **180 ms** including TTS.
- Engineered a modular inference architecture using **MediaPipe, TensorFlow/Keras, and OpenCV**, enabling deployment across **3+ configurations** with automatic fallback and sub-**250 ms** recovery from sensor failures.

TRIDENT — Tactical Disaster Response & Rescue System *(IoT, Robotics)* [**Demo**] [**GitHub**]

- Built an edge-based rescue platform integrating a smart wearable, dashboard, and ROV, streaming **real-time vitals, motion, stress, and GPS data** with alert latency under **1.8 s** and telemetry uptime of **>95%**.
- Designed rule-based and probabilistic scoring to classify victims into **Priority 1–5**, reducing manual triage time by **40%** and improving decision accuracy by **30%**.

DrishtiGuide — Assistive Navigation System *(YOLOv8, IoT)* [**GitHub**]

- Built a real-time multimodal navigation pipeline (audio, haptic, ultrasonic), reducing missed-obstacle events by **∼60%** versus cane-only navigation in field trials.
- Integrated **YOLOv8, OpenCV, and PyTorch** with ESP8266 telemetry, achieving **sub-120 ms inference latency** and **>99%** reliable GPS alert delivery to caregivers.

Skills

Programming Languages: C, Python, Java, HTML, CSS
Frameworks & Libraries: OpenCV, PyTorch, TensorFlow, YOLOv8, ROS 2, NumPy, Pandas, MediaPipe, CVZone
Systems & Platforms: Linux, ROS 2 Middleware, Real-time Data Pipelines
Development Tools: Visual Studio Code, Git, Github
Professional Skills: Software Design, Problem-solving, Debugging, Collaboration, Time Management

Achievements & Leadership

- **SIH'25 Finalist** — advanced to **Top 50 out of 864 teams** internally and **Top 5 out of 500** teams nationally for the Grand Finale. [**Certificate**]
- **3rd Place, Technovation 2025** (Kurukshetra '25, CEG Guindy, Anna University) — ranked **2nd Runner Up out of 50+** teams; ₹ 15,000 award. [**Certificate**]
- **Winner, Innovate X Impact (V-NEST)** — secured **1st Position out of 35** teams; ₹ 5,000 award.
- **Deputy Operations Lead, Team Namo Nirvana (VIT Chennai)** — led **100+ member** teams across **5–6 competitions/semester**, boosting execution efficiency by **35%** and securing **2 sponsors**. [**LOA**]

Languages

English (Fluent) — Hindi (Native)